Math 420: Homework 2

Name:

September 10, 2025

1. Prove that

$$\det\left(\begin{bmatrix} 1 & 2 & 2 & \dots & 2 & 2 \\ 2 & 2 & 2 & \dots & 2 & 2 \\ 2 & 2 & 3 & \dots & 2 & 2 \\ \vdots & \vdots & \vdots & \ddots & 2 & 2 \\ 2 & 2 & 2 & \dots & n-1 & 2 \\ 2 & 2 & 2 & \dots & 2 & n \end{bmatrix}\right) = -2(n-2)!.$$

2. Let U_1, U_2, U_3 be subspaces to a finite dimensional vector space V. Show that

$$\dim(U_1 \cap U_2 \cap U_3) \geq \dim(U_1) + \dim(U_2) + \dim(U_3) - 2\dim(V).$$

3. Let $V = \operatorname{span}\{AB - BA : A, B \in \mathsf{M}_n(\mathbb{R})\}$.

   (a) Show that the function $\operatorname{tr} : \mathsf{M}_n(\mathbb{R}) \to \mathbb{R}$ is a linear transformation.

   (b) Prove that $\dim \ker \operatorname{tr} = n^2 - 1$.

   (c) Prove that $\dim V = n^2 - 1$.